

A radioactive source emits gamma radiation. A detector measures the count rate at two distances.

(a)

Calculate the new count rate at a distance of 0.40 m

The count rate follows the inverse square law:

$$\frac{I_1}{I_2} = \left(\frac{r_2}{r_1}\right)^2$$

Given:

• $I_1 = 1500 \text{ counts/s}$ at $r_1 = 0.20 \text{ m}$

• $r_2 = 0.40 \text{ m}$

$$I_2 = I_1 \left(\frac{r_1}{r_2}\right)^2 = 1500 \left(\frac{0.20}{0.40}\right)^2 = 1500 \times \left(\frac{1}{2}\right)^2 = 1500 \times \frac{1}{4} = 375 \text{ counts/s}$$

(b)

State one assumption made in this calculation

- ✔ The gamma radiation spreads **uniformly in all directions** (i.e. it’s emitted **isotropically**) from a **point source**.
- ✔ No significant **absorption or scattering** occurs between the source and detector (air is treated as transparent).

(c)

Explain why the gamma radiation is suitable for this investigation

- ✔ Gamma radiation is **highly penetrating**, so:
 - It is **minimally absorbed by air**, ensuring accurate inverse square law behaviour.
 - The detector reading primarily reflects **distance**, not energy loss from absorption.

 Revision Tips

- Inverse square law applies to radiation spreading out **spherically** from a point
- If distance doubles, intensity drops to **one quarter**
- Gamma rays are ideal for this because they are **least affected by air or materials** over short distances
- Always consider **assumptions**: point source, no absorption, constant detector efficiency, etc.